

ActInf Textbook Group ~ Cohort 6 ~ Session 8 (Chapter 5, Part 2) ~ 4/8/202

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 8 (Chapter 5, Part 2) ~ 4/8/2024 | 41:34

all right well thank you for that very
interesting prior discussion on on
latter textbook group so I hope that we
can continue from that first few minutes
that we had okay this is the last
recorded
discussion feel free to come next week
for um the other cohorts also unrecorded
discussion again we'll kind of like look
over what has been written think about
what we want to do with textbook group
and everything kind of continuing
so but here's chapter 5 and or like
anything on the first half of the
book like anyone could go any part from
five or first half or we could look at
the questions and look at which ones or
spend a minute and upvote questions that
we're interested in and then look at
which ones have been upvoted a lot but
not
answered for first
time sure question sound
good okay in chapter five anyone can
just people can
be clicking the thumb and then we'll
just go in these or raise your hand and
like go for anything else but
anything in chapter five that anybody
wants to go to
first um table 5.1 with the
neurotransmitters um I think it's page
98 yep that's the
one um
so is what being stated here pretty much

saying that you dopamine is increasing
the Precision of you know the the brains
uh you know I guess what the bra
understands of of
policies so it's saying you these
policies are more precise if you know
the uh a newon or a group of new on is
getting more to
mean yeah so to kind of like pull back
on the table this is going to be linking
specific molecules and roles they play
in mechanisms in the brain to specific
variables in generative
models and then the table is like
describing well here's some sheer
anatomical
facts these are empirical measurements
and then
also describing and citing some studies
that might have studied um the
relationship between like the activity
level or the amount or the signal of
some neurotransmitter to some aspect of
a generative model and then that could
be more correlative
evidence like size of a brain region or
um changes in the functional
fmri um okay or it can be more causitive
with like CH giving a manipulation and
then showing that the two things are
changing together so it doesn't mean
it's the only function it does or its
whole role or anything but it's just
saying these parameters are being being
identified are not just like weird
statistical artifacts they map on to
like natural components of
behavior so it's proposing there some

connection between your know dopamine
and the policies that the brain is
building there's to to really look at
this in the dopamine case thankfully
which is actually cool because it's most
most things like the kind of case for
pluralism has not always been written
out this is okay we'll just go in here
but in
this this paper they specifically are
looking at
the meso cortic olivary dopamine so this
is something like what is shown in the
um figures in chapter 5 with dopamine
controlling the balance between the
habit-based policy selection
and the expected free energy based
policy
selection so that's kind of that's the
computational function that is being
posited in the model that's in the
chapter and it's what it's being
identified on for um in the
table and
then this paper's like philosophers
biology and saying well people have
modeled dopamine in terms of the homia
the salience and the reward prediction
error basically free
energy and they say well there's already
evidence to support pluralism for that
system but you could make probably
analogous Arguments for like every brain
region every Gene
Etc but that's the whole thing with the
maps in the
territories it's like okay it's a subway
map of this city but it's not an

everything

map

okay okay any other random part of five
or one of these questions someone wants
to think

about sure just to uh to follow up on
that last uh question and point really
quickly um I think there's a good
example this is in an older paper I
think it's called um dopamine reward
learning and active
inference and um it looks at a lot of
aspects of like dopaminergic uh function
in the brain you know as interpreted
through active inference models one one

I think is kind of particularly
interesting is um freezing of gate in
Parkinson's I'm not sure if folks are
familiar with that but um long story
short people with Parkinson sometimes
get this thing where like you get up
from a park bench and you try to you
know walk across the lawn and and the
person will kind of freeze and there
haven't been really great explanations
about what that's about and like why it
happens um but one of the things that I
guess uh kind of emerges from this paper
if I recall right is the idea that there
are basically a bunch of different
action policies you know that you can
take to navigate the park right you can
cross the space in a number of different
ways and basically if you have low
dopamine levels like people do in
Parkinson's then you can have a low
potential Precision that's assigned to
any one of those given action plans and

so with no ability to sort of assign
High Precision to one plan or another
you have kind of this series of low
confid confence plans and the Brain
can't kind of like select one on that
basis and so you kind of just lock up um
and there are like techniques that
people with Parkinson's learn to do
where they learn to like intentionally
choose like okay I'm going to walk in
sort of a crescent pattern around this
area so they basically like make a
different decision that kind of like
jolts the system out of um out of the
kind of lock state that it gets caught
in and you know ultimately I don't know
about like are there experimental
results that really support this kind of
thing I'm not totally sure but I at
least I found it to be a really
interesting way to think about this kind
of thing and I haven't found
another um another sort of explanation
that I think is is sort of equally as as
elegant that's kind of interesting it's
like from an that perspective it's like
using the brain the brain's plasticity
to uh overcome um just errors in its own
system as it
ages but great yeah
comments that just in terms of like
research agenda this is like a
preliminary stage where you can take the
extent data and then make a model that
fits the variance patterns better Andor
has fewer parameters so it's a better
model just by the normal
Criterion so that's a

computational one and then with that model which can be argued to be like better or simpler than Alternatives it's like well that's the leading candidate to then design experiments that would differentiate and maybe even let the model um show like new predictions or explanations but it's interesting like it's

2015 and it's a very similar figure to figure

4.3 with some more details

including about that gamma in the table also this is kind of interesting like without the textbook seven years before the

textbook every

paper has a lot of the background equations whereas now it's clear which parts are referenceable like to just the textbook and then which parts still might be highlighted in the paper like the focus on the

policy and the RO devopment

policy yeah interesting

paper maybe it's SED in

here maybe

it's that's this one

one or maybe it's another one by the same authors

right or no this is only there three of them

anyways yeah any other just random thought or or upload questions if you have the Koda

up

this is kind of like

the theor theorizing around like

relationship between the fun fun and the
anatomy for the brain or the function
and the genome like there's kind of a
simpler one gene one protein one
function or like one anatomical region
one Function One computational role but
of course that's not how any biological
system

is so it's kind of a straw hypothesis

[Music]

to

merely

disprove because there's so few things
that have are with such straightforward
causation but those are the interesting
ones in some ways but they're also not
representative of other

cases

here's one Mike Len paper to more recent
and Josh

bongard I'll just copy it in but it kind
of talks about like the overloaded
nature of the biological
systems

whereas a computational system you could
engineer it to not be

overloaded like you could make a which
is kind of part of the reason why they
don't capture those biological

Dynamics

because the basing graph like it's just
kind of not

overloaded among other things

thank you

Cheryl

welcome

yeah one book that speaks to um what a
little bit of what you describe like the

amigdala and the like social events is
um this book by Helen
Lino about some charged slash relevant
topics and how
pluralism

gets complex but also like very
important you know Billboards it's in
our DNA and all those kinds of things
those are part of the cultural milieu that
the

scientific facts even to the extent that
they're empirical factual those are
mobilized

and that's the whole
discussion

it

and she talks about complex systems and
like developmental systems theory as a
way to link different scales of analysis
so it's kind of like Mike Levan's work
like if you could look at the
morphology of organism growing and then
over slower time scales you have like
structural

factors but they're happening at
different time scale so neither like the
development of the organism or like
structural slower

factors could explain what the other one
does there's there's a lot more I think
in that direction that would be awesome
to explore on like kind of some of the
broader tangents like Oli and that other
people bring um up with like
very contextual but very rich points
about like reductionism and
holism free will all these other more
psychological things too but also a lot

of

scientific um

epistem slash discussion

Susan um yeah you know it's

mentioned kind of um

offand um in a number of places as far

as the um cybernetics and the valuable

systems model and yeah I it'd be hard

for me to understand uh this if I didn't

have a a good understanding of

cybernetics um and yeah there's a to to

your point

um um about it being reductionist Cheryl

Cheryl there's uh yeah I you you do kind

of hear that in terms of how people are

approaching it um and but the same thing

I think is is true of cybernetics the

original cybernetics was very mechanical

um but they've since you know embedded

it and Infused it with h with more of a

of a Nature's approach so yeah it's it's

kind of everything's evolving so you

know if we look at it through a static

lens we're kind of already GNA be stuck

because it's all moving it's all flowing

um and that's kind of where we're trying

to grasp something but it's flowing away

from us and which I think is is

fascinating about the topic but I also

think that

understanding the different levels of

ambiguity and

abstraction it's hard to you to grasp

with that in terms of how do you contain

the conversation to go deep when there's

so many Avenues to jump on um in fact a

question that I had um in terms of

Markoff blankets um which uh it might

sound like I'm going off on a tangent
but there is actually connection is is
there in Markoff blankets um what would
be the uh the interdependencies how
would you how do you
describe the
interdependencies or are markoff
blankets
essentially boundary
interdependencies anyone give a thought
on
that okay one way to think about it
maybe narrower or whatever but since the
base graph is a causal graph it's a map
determines complexity around what is a
determinance but that's how variables
determine the influence affect each
other that's what the edge on the
causal graph is so what's there are the
determines but what isn't there which is
what makes the internal external States
separated if they had an edge between
them that would be then that would
change the shape of the
graph another thought related to this is
like this this may be a very coarse
first approximation but just to to
reduce some of it to
categories a lot of the questions relate
to questions about modeling at
all as well as some other very general
topics like predicting at all acting at
all and then there's also finer scale
questions about how active inference
formalisms related to like reinforcement
learning so that's those are very
different
contexts that many people in textbook

group are are totally more than Adept at
so that's part of the fun also though
that may be harder contact shifting but
it's also real contact
shifting so it's not to be like
removed but knowing how to approach more
of this like what can numbers and maps
have to do with systems at
all which is very helpful to have people
with some shared background
on and then
also some will then get to more of a
level of resolution of differentiating
active inference versus
adjacencies in like perceptual control
theory reinforcement learning everything
that's basically discussed in chapter 10
like from a theory perspective and then
also alternative mathematical models for
like the mammalian nervous system in
chapter 5 because it's easy to be be
skeptical and bring up many General
points about the models presented in
chapter
5 and it's also relevant if one really
wants to go into it
to contrast not having a model versus
not having a model but what are two
models for the computational role of
dopamine and then which one of those are
more plausible
and tractable and general and so on but
that's a very different discussion than
should we make a map of policy selection
at
all or what could ever be said about a
biological organism with
math this is like one of those kind of

philosophy quotes that that ends up
being very influential for a very long
time this quote by Kant there will never
be Newton of the blade of
grass people debated still
debate

very good discussions very great points
and things that I think a lot of people
could enter into though very deep
within philosophy texts and kind of
Niche but this sounds
awesome

I mean like even just the title of the
paper we can read later but I mean this
is kind

of this is philosophical context for any
mathematical work like

ACM

okay

any other part of five or the first part
of the book or like any other
thoughts or I mean what

else I had a question about uh the
degree to

which their description of

uh the math being implemented in the
actual neural

system uh implies something about like
what I would normally think of as
modeling assumptions about

parametricity um being sort of baked
into the mechanics um so in particular
right there's like all this focus on uh
precision and I understand why that's
important um but like in other contexts
like there

are

other statistics later moments that we

might care about that like are important to understanding like the shape of a distribution for example um and it normally I'm used to seeing like when when like you know there's an assumption in some paper that like something is gaussian uh that that is like a modeling convenience rather than a statement about like a biological reality I I guess when when Precision is like hypothesized to be like an actual thing that's communicated like in in particular systems of the brain does that kind of imply that like particular behaviors can't learn things that are dependent on like the shapes of tails of the distributions that are being communicated about rather than just like their broadness versus or uncertainty around an estimate yeah a lot of great questions in there um few different pieces there's like this discussion of the support early in the textbook it's kind of like why and then it's it's kind of like what you brought up like but then the Gan goes off to very low so does that mean there's like an infinitely low probability of learning something infinitely rare and like all all these different questions so then like that is the statistical question is like how should it reflect if it's a gaussian situation then that one parameter will do fine capture all variants but then if it's anything

different than that which is basically
everything then um and then that also
reminds me of
um a discussion in this paper or uh SL
Liv

stream and so here they they were
summarizing networks not like just with
a mean and variance but summarizing them
with more of like a latent space that
could be larger um or smaller so it's
basically what a variational auto
encoder is but they were doing that as
like this kind of meta representation it
was like it was like very interesting
but he shared um but this is basically
the tradeoff if you only have two
parameters if you only have one
parameter

to distill the information projected
down to then some common choices are
like the median the mean arithmetic or
geometric mean um or like the waiting
time for a single parameter model then
with two parameters mean and variance
like the measure of central tendency in
the first one movement and then the more
parameters you have the more expressive
you can

be but that's the whole question with
these large parameter models
so you can make and and what
uh what riota discussed in in this was
actually like and kind of to the
Consciousness point was like so the
global workspace like the ignition
across
different sensory modalities so that's
the kind of binding of the

Consciousness and previously it was studied like does it happen or not or like was the shape perceived or not was the stimuli perceived or not as a kind of conscious to not conscious but then once you're in the space of things that have been perceived then there's a need for richer differentiation because they don't just differ in that one is not the other so then he talked about how you can have richer representations than than all or none

another um more statistical answer is uh there's probably a lot of this is you can compose gaussian distributions to approximate any distribution like basically so it's not just two parameters in the actual papers but that's kind of the minimal explainable component like SPM like in the end it's going to result in this map of the brain with statistical with one value per voxel which is like basically the statistical test on whether it's different but there's a ton of parameters up stream of distilling down to one statistic there's very active SPM discussions like they have a mailing list and there's trainings and things like

this I don't know if anyone has done the text has done this SPM work but it it may be relevant for the textbook SPM deals less with the action

selection and philosophy and more with
the statistics
sanj's textbook is probably more direct
way to learn the statistics
specifically however this is still
informative
statistics just to orient myself SPM is
the thing that first made frisan
famous yes so SPM is a mat lab toolbox
for modeling the functional
fmri Dynamic time series data and like
including other neural sensors so
basically Sensor Fusion in
neuroimaging and hence being Mega cited
slash being the bread and butter and the
method of
the priston
lab then early in the
2000s this kind of general just take
formal approaches to perception
cognition
action also started to combine with some
of the first principles and the physics
of cognition and sentience and things
that so this type of a statistical
framework is not focused on action on
first past it's just modeling
observables the sensor data and then the
hidden state which is the neural
activity so it's like a a matrix and a b
Matrix in the textbook no policy
selection it just kind of or or it's
kind of like um
the chap figure 9.1 like you're viewing
the outside so it's like what kind of
modeling can you do if all you have is
neuroimaging data and behavioral data
and information on the people like their

you know other details about
them and it's very use oriented like all
the citations to
this let's
see

SPM

SPM 2010 paper being kind of a landmark
Fe

paper SPM DCM Dynamic causal modeling
looking at the the dynamical causal
relationships so then like to test
whether attention is engaged in a given
moment it's like okay well the
attentional network connects these nodes
and then when it's not attentive it
disconnects these nodes

and then we're going to test whether
this or that model explains this or that
behavior very foundational
statistical work about like correcting
for noise and variance patterns in fmri
like slower and faster um noise like
Varian and

deviations then also first and all have
the Clinical

Psychology background so then using the
mat lab toolkit they have a lot of math
papers on the then the toolkit and then
also like looking at
specific like the stuff that we saw in
the table 5.1 with like

specific brain regions in specific
behaviors and many people who still do
fmri mean they use this toolkit

let's see the first where AC okay two
first active

inference firston

pulo kind

of this is probably similar to the
textbook about five years earlier
pretty
interesting but that's all every field
has more papers to read than SL order
what to do them in and all of
this

but it does really emphasize that more
of a path-based approach is
relevant and also augmented with mega
new tools for like processing all of it
and so

Susan yes can you can you say anything
about the um the page on active
inference model recipe that's pretty
cool it's under chapter
six this is kind of their General layout
of the book chapter 1 through 5
the epistemic and then chapter 6
through1 the um pragmatic okay that
specific

page okay yeah there it is this may have
been gone through this is just taking
the questions that were in the
chapter and then that's the ones that
are basically just in explained in the
subsections okay we also use a language
model to ask for more detail on all the
sections and then this was um I believe
with Michael Carl who later presented
some of the work on a stream actually it
was very cool about the data set that
they had which was related to
translators actions like with ey gaze so
where they were looking in the text and
then what they were typing and deleting
and
then just doing descriptive statistics

like of course you could categorize
different categories of
translators that would be like you could
use a simpler statistical approach or
like a deep learning approach because
you have the data or and or what does it
look like to make an active inference
cognitive model that describes the
generative
process of the translation
process which is a complement to the
descriptive
statistics like generating synthetic
data for models that could then be
descriptively um describing that output
yeah
so is this
assuming that you there is an observer
and an observed
or in terms of you know because it's
like the question of whoop um you know
what must be
observed could that
be U could that be you know self
modeling yeah I'm not GNA know what I'm
observing other than what I'm observing
if I'm the Observer versus the
subject yeah it's it's a very
interesting question like definitely s
have some private access to certain
things but in the case of the
Translating
like Professor Carl was he it was his
laboratory so he knew exactly what the
stimuli were they were the observations
for the translators the strings of texts
and then he knew what measurements the
actions of the translator were being

observed Obed and this is the kind of
like in SPM it was initially with fmri
with the MRI imaging time series and
then also they're able to combine that
with Meg and EEG and and probably other
neuroimaging techniques it doesn't even
matter what the acronyms are the point
is you can have different sensor sets
and merge sensor sets with deeper and
deeper generative
models so if they also had the
translators like
rate that would be the question okay how
do we bring the heart rate
in but that doesn't change the
affordances but then let's just say also
you start looking at their foot tapping
then that would be like okay now that's
a new affordance to model so that's kind
of the interoperability of the active
inference
models with deep learning some of these
models do have
multimodal
capacities like the ones that can take
in text or images and then output like
text or
images so it's not that other models
can't do
that it's just there's a lot of other
things to say about that like the
interpretability or the reliability of
adding a new
modality all these different
features thank you but to make a
pedagogical fun exploratory model then
those considerations are not going to be
hindrances so hence chapter six I think

that would be something fun like that's kind of the pacing that we can figure out not necessarily two weeks on six but as as I mean Susan and others no like once we start getting it down in this basically chapter 6 format then we're in feedback with what we have it's easier to communicate it easier to build on it also the technology for applying it given even like this level of detail are far more advanced than they were like one or two or more years ago so it's not like exactly like you could just copy in this little and get a script today but it's not so far away but it wouldn't even matter if it were there would still be going deeper and being more expressive and having in our own generative models more of an understanding and yet also like you can use the GPU and deep learning without knowing all the GPU like Linux kernel details so that's like levels of abstraction in the modeling those are just different experiences somebody wanting to just run the software I mean look at the Port run the software run the kind of fully investigable version understand the minimal unit like understanding the minimal unit of a neural network those kinds of things well next week it'll be the last part of interval coming to this or the other time and then we'll see how people want to um continue six

and like also we could explore like how
do we want to interact it with seven you
know different things we could talk
about
okay thank
you take care everyone